

MACS 40550: Final Project

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Research Question

Under what conditions does social observation transform initially indifferent agents into active participants in scarcity purchasing, and when does the cascade of rising demand collapse before supply is exhausted?

Literature Review and Framing

The relationship between scarcity and perceived value is well-established in the literature. Worchel et al. (1975) demonstrated in a classic experiment that limiting the availability of an object increases its perceived desirability independently of any change in its intrinsic quality, the finding since replicated and extended across numerous consumer contexts. Lynn (1991) conducted a quantitative review of the commodity theory literature and confirmed that scarcity effects on value are robust across product types and cultural settings. These findings establish that limited availability is not merely a constraint on supply but an active signal that shapes how consumers evaluate goods.

Existing computational methods of consumer behavior tend to treat demand as exogenous, assuming that the number of people who want a product is determined prior to and independently of the social process by which purchasing unfolds (Janssen and Jager, 2001). This assumption becomes problematic when we consider platforms and environments in which consumers can observe each other's interest and behavior in real time, such as online limited-edition drops, live-streamed product launches or group purchasing communities. In these settings, scarcity and social observation operate together. The visible fact that supply is limited motivates consumers to monitor what others are doing, and observed purchasing behavior in turn amplifies individual desire. Demand is therefore not fixed prior to the social process but is partly constructed through it.

There are two pieces of literature directly related to the possibility. Granovetter (1978) The threshold model of collective behavior provides a framework for understanding how individual-level heterogeneity in

decision thresholds can produce aggregate patterns of adoption or non-adoption that are highly sensitive to initial conditions. In the model, each individual has a threshold representing the number of others who must act before they are willing to act themselves. The resulting dynamics produce tipping points and cascade failures that are not predictable from the distribution of individual preferences alone. Bikhchandani et al. (1992) shows the complementary mechanism through informational cascade theory, showing that rational individuals may choose to disregard their own private information and imitate others when the observed behavior of predecessors is sufficiently informative. Together, these frameworks predict that social observation can amplify or suppress demand in ways that are highly nonlinear and path-dependent.

The gap that I want to fill is a model in which demand itself as an emergent outcome rather than the starting condition, shaped by the weighted influence of neighbors whose interest is observable and tie strength varies. While existing models tend to isolate one dimension of this process at a time, the model integrates scarcity, social contagion, and network tie strength within a single framework to examine how demand emerges from their interaction.

Mechanism Description

The underlying social mechanism I model is one of demand contagion moderated by scarcity. Each agent begins with their own private interest level reflect their baseline disposition toward the product, which may be low or high depending on individual heterogeneity. As agents observe neighbors expressing their interest or actively purchasing, they update their own interest upward through a social learning process. Conversely, neighbors who visibility fail to obtain the product send a discouraging signal that suppresses their interest. When an agent's interest level crosses their personal threshold, they join the purchasing queue.

Scarcity plays two distinct roles in this mechanism. First, limited availability serves as a social cue that amplifies urgency: the visible fact that others are competing for a scarce good accelerates interest updating among observers, consistent with the reactance and commodity theory mechanisms documented by Worchel et al. (1975) and Lynn (1991). Second, the supply depletion creates a structural constraint: agents who are influenced to act late may find the good no longer available, and their visible failure sends a discouraging signal that suppresses the interest of agents who have not yet decided. The first role scarcity drives demand upward, and the second role can arrest or reverse the cascade. Whether the cascade reaches most of the population or collapses prematurely depends on how quickly contagion spreads relative to how quickly supply is depleted.

The critical feature of this mechanism is that the tie strength between agent is not uniform. Each connection is independently classified as either a strong tie (friend/family) or a weak tie (stranger). Strong ties carry full influence weight while weak ties carry substantially reduced weight. This means that the model holds the total number of connections constant while varying the proportion that carry high versus low influence,

allowing use to ask whether tie quality independently shapes cascade outcomes beyond what tie quantity alone would predict. The model captures the intuition that peer behavior matters more when the peers are people we trust.

Once an agent fails to obtain the good, they do not attempt again. This reflects the structure of real limited-release purchasing environments, such as online product drops or lottery-based sales, where the purchasing window closes at a fixed point and a failed attempt is permanent rather than repeatable.

Description of Model

The model is a network-based spatial structure. Agents are placed on nodes of either Erdős–Rényi random graph or a Watts-Strogatz small-world graph, selectable by the users. The Erdős–Rényi graph creates connections with uniform probability, producing a network without strong local clustering. The Watts-Strogatz graph starts with a ring lattice in which each node is connected to its nearest neighbors and then rewires each edge independently with probability $p = 0.1$, producing a locally clustered network with some long-range shortcuts that approximates the structure of real social networks more closely.

Each agent is initialized with two key attributes. The first is an interest level drawn from Gaussian distribution: most agent begin with low interest (mean=0.15, standard deviation=0.1), while a small proportion as enthusiasts or influencers that begin with high interest (mean=0.85, standard deviation=0.08). Influencers are defined as the highest-degree nodes in the network, reflecting the empirical observation that highly connected individuals are disproportionately likely to be early adopters and opinion leaders. The second attribute is a personal threshold drawn from Gaussian distribution with a user-specified mean and standard deviation, capturing heterogeneity in how much social proof different agents require before acting. Both the mean and standard deviation of the threshold distribution are exposed as parameters, allowing the user to shift the population-level susceptibility and vary how much agents differ from one another.

Each edge in the network is independently labeled as either strong tie or weak tie with probability equal to the friend ratio parameter. Strong ties carry a weight of 1.0 in the influence calculation and weak ties carry a weight of 0.2. At each step, non-terminal agents capture a weighted average of their neighbors' signals, where a neighbor in ACTIVE or SUCCESSFUL state contributes a signal of 1.0, FAILED neighbors contribute -0.2, and all others contribute their raw interest level. The agents' interest is updated toward the weighted signal at fixed learning rate of 0.3, which reflects a moderate updating speed and was chosen to produce gradual cascade dynamics rather than immediate full adoption. The discrete state is updated accordingly.

There are five states in the agent cycle. INDIFFERENT agents have an interest level below 0.3 and have not yet been sufficiently influenced. INTERESTED agents have an interest level between 0.3 and their personal

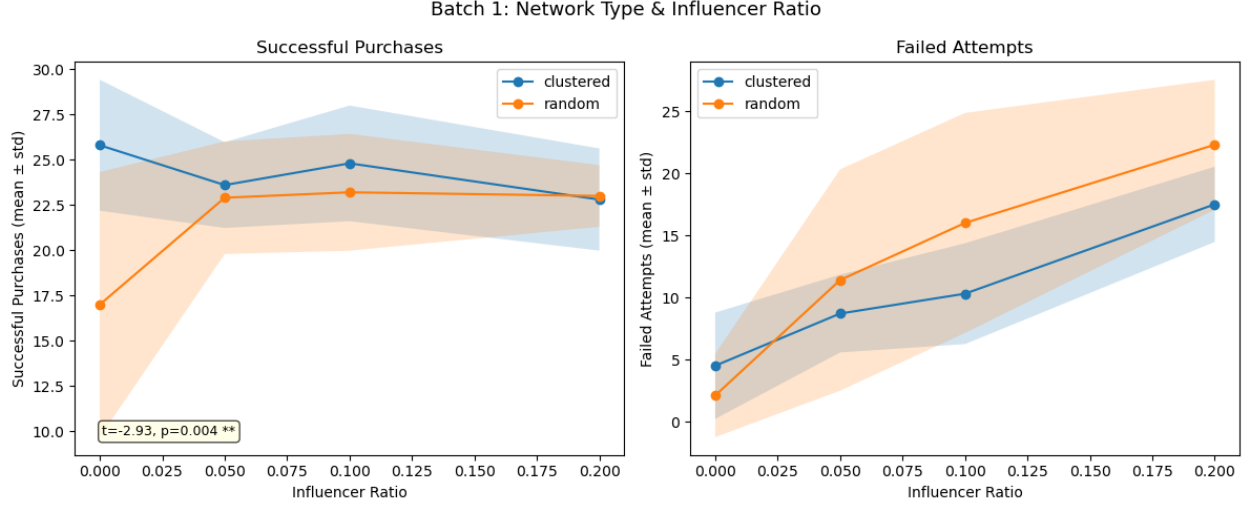
threshold. ACTIVE agents have crossed their threshold and are attempting to purchase in the current step. SUCCESSFUL agents have obtained the product and no longer act, though they continue to emit a positive signal to neighbors. FAILED agents attempted to purchase but found the supply exhausted; they emit a small negative signal and no longer act. Agents step in a random shuffled order each round, implementing a first-come-first-served allocation that is equivalent to a random queue.

Supply is initialized at a user-specified level and decremented by one each time an agent successfully purchases. At fixed intervals determined by the restock interval parameter, a proportion of the initial supply is restored that up to the original level. The stopping rule happens when no agents remain in the ACTIVE or INTERESTED state, meaning the cascade has fully resolved. Key outputs collected by the DataCollector at each step include the counts of agents in each state, the cascade size (ACTIVE plus SUCCESSFUL), and the remaining supply level.

Analysis

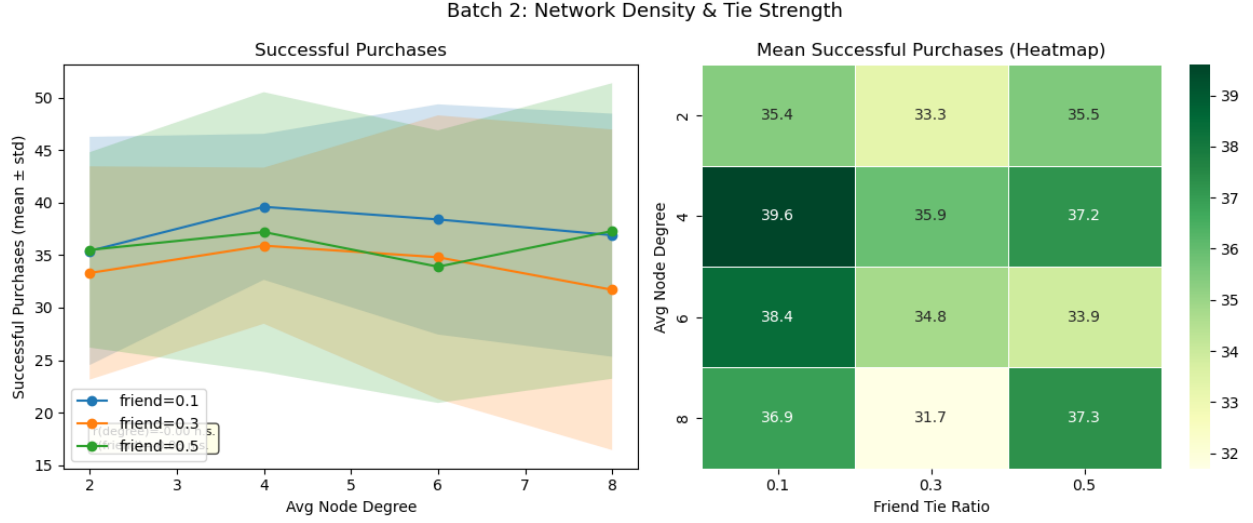
I conducted four batch runs to explore conditions under which demand cascades amplify or collapse. Each parameter combination was repeated ten times with different random seeds.

Batch Run 1: Network Structure and Influencer Ratio



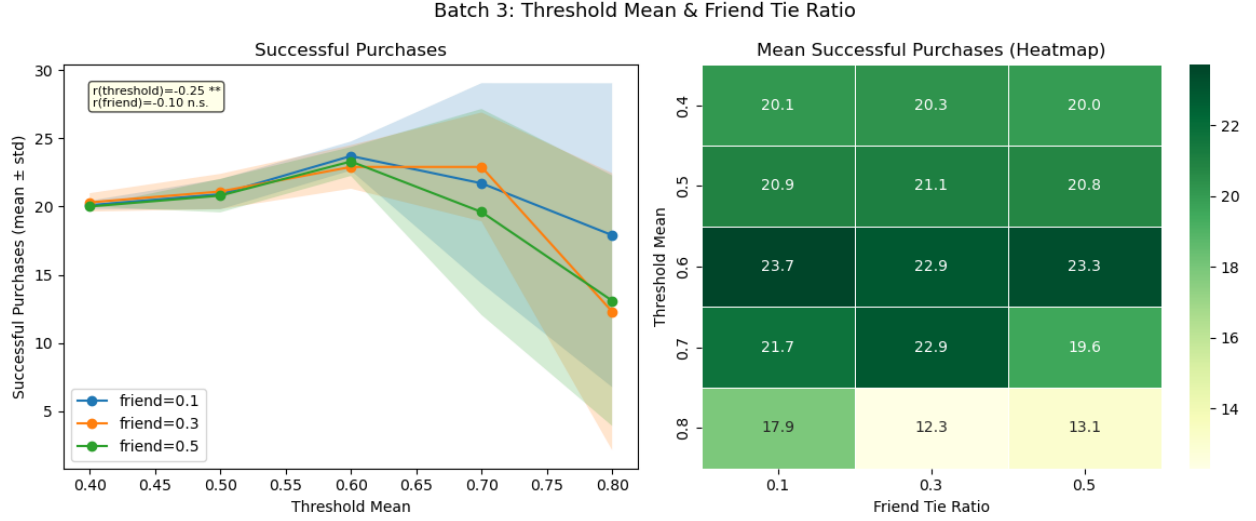
The first batch run varied network type (random versus clustered) and influencer ratio (0%, 5%, 10%, 20%) while holding all other parameters constant (supply = 20, restock rate = 0.2, threshold mean = 0.6, friend ratio = 0.3). The result shows that clustered networks had significantly more successful purchases overall ($M = 24.2$) compared to random networks ($M = 21.5, t = -2.93, p = 0.004$). This difference was most in the complete absence of influences that without any high-interest seed agents, the clustered network achieved the mean of 25.8 successful purchases compared to only 17 in the random network ($t = -3.41, p = 0.003$). The pattern is consistent with the threshold model logic of Granovetter (1978): early adopters being densely connected to each other in a clustered network, which allows social proof to accumulate locally before spreading outward. Once enough agents within a cluster cross their threshold, the entire cluster tends to activate in sequence. In the random network, the same number of early adopters is spread thinly across the network, and no single neighborhood accumulates enough signal to trigger a local cascade. As the influencer ratio increased, the gap between network types diminished and became non-significant at ratios of 5% and above. This suggests that influencers effectively substitute for structural clustering by broadcasting high-interest signals to many weakly connected agents simultaneously. Notably, the right panel reveals that random networks accumulate substantially more failed purchase attempts at higher influencer ratios, indicating influencer-driven demand in sparse networks outpaces supply availability, producing cascade overshoot rather than cascade completion.

Batch Run 2: Network Density and Tie Strength



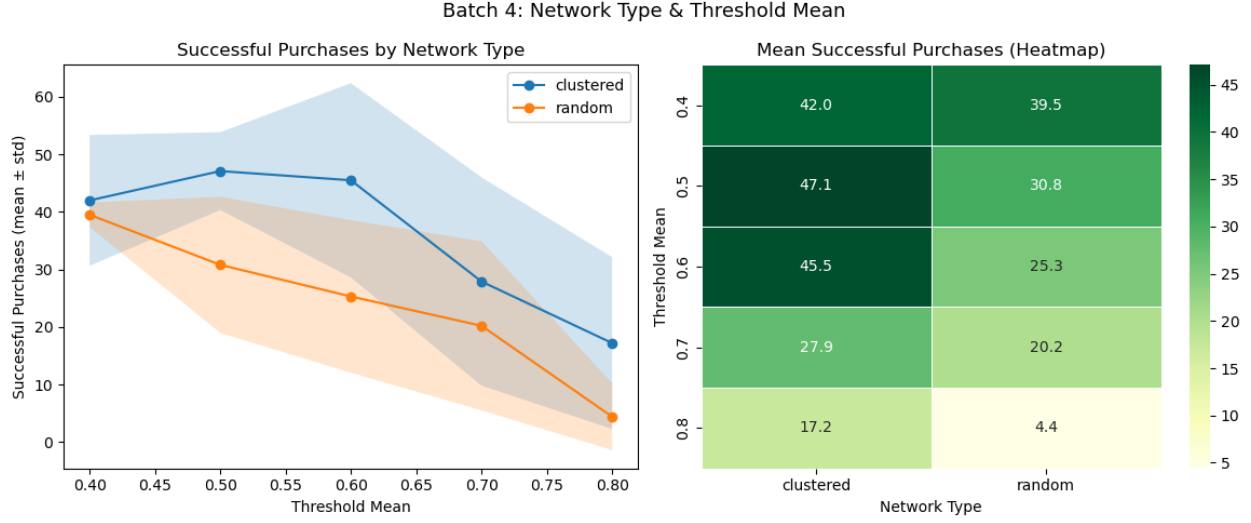
The second batch run varied average node degree (2, 4, 6, 8) and friend tie ratio (10%, 30%, 50%) while holding all other parameters constant at supply = 40, threshold mean = 0.6, and influencer ratio = 5%. Supply was set generously to remove it as a bottleneck, so that any observed differences could be attributed solely to network density and tie composition. Average node degree was varied from 2 to 8 to span the range from near-isolated agents to a densely connected network, testing whether tie strength becomes more consequential when each agent has fewer neighbors to observe. If strong ties matter, their effect should be most visible in sparse networks where each connection carries more weight in the influence calculation. Neither average node degree nor friend tie ratio showed a significant correlation with successful purchases ($r = -0.002, p = 0.986$ and $r = -0.058, p = 0.528$ respectively), and pairwise t-tests comparing low and high friend ratios at every density level were uniformly non-significant. The null result held even at the lowest density tested (average degree = 2), where each agent observes only two neighbors on average and the influence of any single tie is maximally consequential relative to denser conditions. A possible mechanism is that the model's iterative updating allows weak-tie signals to accumulate across multiple steps: even a small per-step nudge from a stranger compounds over time into sufficient influence to cross a threshold, removing the need for the stronger per-step push that a friend tie would provide. This finding is further confirmed by Batch 3 below, which shows the same null result for friend ratio at the default density level. Together, the two batch runs suggest that tie strength is structurally irrelevant within this model's parameter space, not merely at moderate density but across the full range explored, and that interventions aimed at strengthening relationship quality within a consumer community are unlikely to substantially alter cascade outcomes.

Batch Run 3: Threshold Mean and Strong Tie Ratio



The third batch run varied the population threshold mean (0.4 to 0.8) and the proportion of strong ties (10%, 30%, 50%) while holding supply = 20, restock rate = 0.2, network type = random, and influencer ratio = 5% constant. Threshold mean was varied across this range to span the full spectrum from a highly susceptible population to one that is nearly resistant to social influence, with the expectation that tie strength would matter more when agents require stronger signals to act. Friend tie ratio showed no significant correlation with successful purchases ($r = -0.1, p = 0.223$), and pairwise t-tests comparing low and high friend ratios at every threshold level were uniformly non-significant. From the heatmap, successful purchase counts are remarkably stable across all combinations of threshold mean and friend ratio between 0.4 and 0.7, ranging from approximately 20 to 24. This stability finding is surprising because strong tie theory would predict that a higher proportion of friend ties should amplify social influence and facilitate cascade spread. One possible explanation is that in a network with 100 agents and average degree 4, even weak-tie signals accumulate sufficiently across multiple steps to push agents across their thresholds; it renders the strength of any individual tie less consequential. Threshold mean showed a significant negative correlation with successful purchases ($r = -0.253, p = 0.002$), but this effect was largely driven by the highest threshold level: at threshold mean = 0.8, mean successful purchases fell to approximately 13 to 18 across all friend ratio conditions, while all lower threshold levels produced relatively similar outcomes. The results show the threshold ceiling effect, where moderate variation in how much social proof people require does not substantially change aggregate cascade outcomes, but a sufficiently skeptical population can resist the cascade almost entirely. To test whether this null result was specific to the default network density, Batch 2 varied average node degree from 2 to 8 while varying friend ratio simultaneously. The result remained uniformly non-significant across all density levels, suggesting that tie strength is structurally irrelevant within this model's parameter space regardless of network density.

Batch Run 4: Network Type and Threshold Mean



The fourth batch run varied network type (random versus clustered) and the threshold mean (0.4 to 0.8) while holding supply = 40, restock rate = 0.2, and friend ratio = 0.3 constant, with no influencers present, in order to isolate how network structure interacts with population-level susceptibility when there are no designated seed agents. Threshold mean was varied across the same range as Batch 3 to allow direct comparison, and influencers were removed to ensure that any observed differences between network types reflect structural properties alone rather than the amplifying effect of high-degree seed nodes. Threshold mean showed a strong negative correlation with successful purchases overall ($r = -0.598, p < 0.001$). The clustered network consistently outperformed the random network across threshold levels, but the gap was not uniform: at threshold = 0.4, the two network types produced similar outcomes ($M = 42.0$ versus $M = 39.5, n.s.$) because the low threshold allowed most agents to be influenced regardless of network structure. At threshold = 0.5 and 0.6, the difference was largest and most significant (clustered $M = 47.1$ versus random $M = 30.8, t = -3.79, p = 0.001$; clustered $M = 45.5$ versus random $M = 25.3, t = -2.98, p = 0.008$). At threshold = 0.8, the random network cascade collapsed almost entirely ($M = 4.4$) while the clustered network maintained partial cascade completion ($M = 17.2, t = -2.53, p = 0.021$). This pattern suggests that the clustered network’s advantage is not merely additive but amplified under harder conditions: local clustering provides a buffer against high-threshold populations by allowing social proof to accumulate within tight communities even when the broader network signal is insufficient. The random network, by contrast, cannot generate the local density of signal needed to push skeptical agents across high thresholds, and its cascade collapses as threshold increases.

Discussion and Conclusion

The model demonstrates that demand in scarcity environments is not a fixed individual attribute but an emergent collective outcome shaped by network structure, tie strength, threshold heterogeneity, and supply dynamics. The most substantively important finding is that clustered networks produce more complete cascades than random networks when influencer density is low, because local clustering allows social proof to accumulate within tight groups before spreading outward. This finding has direct relevance to the design of limited-edition products, where seeding a clustered community with a small number of enthusiastic adopters may be more effective than broadcasting broadly across a sparse random network. Batch 4 further showed that this network structure advantage is amplified rather than diminished under harder conditions: as population threshold increased, the clustered network maintained substantially higher cascade completion while the random network collapsed, suggesting that structural clustering provides resilience that becomes most valuable precisely when social contagion faces the most resistance.

The null finding on friend tie ratio is worth noting as a substantive result. Batch 2 tested whether this result was specific to the default network density by varying average node degree from 2 to 8, and found that tie strength remained non-significant across all density levels. This suggests that in this model, the iterative accumulation of weak-tie signals across multiple steps is sufficient to substitute for stronger ties, and that interventions aimed at strengthening relationship quality within consumer communities are unlikely to substantially alter cascade outcomes.

This model contributes to the existing literature by providing an agent-based operationalization of demand as an endogenous social process, a perspective absent from both the scarcity psychology literature and the informational cascade literature. This model opens empirical questions about where real-world social networks sit in the parameter space explored here, and whether interventions targeting network structure or influencer seeding could facilitate or disrupt demand cascades in socially relevant contexts.

External Tool Statement

I found literature through Google Scholar searches using terms related to my idea such as scarcity, threshold models, consumer behavior, and so on. Since I am working on a network model, I structured my code based on network-based examples discussed in class and the official Mesa documentation, including the NetworkGrid API and batch run implementation. I used Claude to clarify how `**kwargs` syntax works in the batch run helper function, to determine that increasing the number of repetitions per parameter combination produces more stable and reliable means.

Link to GitHub Repo: https://github.com/anniek8/MACS40550_ABM

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